Module 06: Learning in Robotics — Adaptive Control

Learning Objectives

1. Define **adaptive control** as the real-time updating of the robotic generative model (system parameters and control gains) during operation.
2. Analyze how **online system identification, gain scheduling, and reinforcement learning** implement Active Inference learning in control systems.
3. Apply adaptive control to achieve robust performance despite **model uncertainty, wear, and changing operating conditions**.

Introduction

No robot model is perfect. Friction characteristics change as bearings wear. Payload properties vary from task to task. Environmental conditions shift — temperature affects motor performance, humidity affects material properties, and surface conditions change with weather. Adaptive control enables the robot to learn and compensate for these variations in real time.

Key Concepts

1. Online System Identification

Online system identification continuously updates the generative model's parameters during operation:

- The robot compares its predicted state trajectory (from the B matrix) to the observed trajectory (from sensors)
- Persistent prediction errors indicate model inaccuracy — the B matrix parameters are updated using recursive least squares, gradient descent, or Bayesian updating
- As the model converges, prediction errors decrease and control performance improves

This is precisely the Active Inference learning loop: observe → predict → compare → update. The learning rate (how aggressively to update) corresponds to the precision on the model parameters — uncertain parameters (low precision) update quickly; confident parameters (high precision) update slowly.

2. Gain Scheduling

Gain scheduling adapts the controller's precision parameters based on the operating regime:

- A helicopter operates differently at sea level vs. high altitude — air density changes affect rotor dynamics. The control gains (precision parameters) must be scheduled based on altitude.
- A robotic arm behaves differently when carrying a heavy load vs. moving freely — the inertia matrix changes, requiring different gain profiles.
- Gain scheduling is the engineering implementation of **context-dependent precision modulation** — the same concept that explains how biological organisms adjust sensory gain based on environmental context.

3. Reinforcement Learning for Control

Reinforcement Learning (RL) can be reinterpreted through Active Inference:

- The **reward function** maps to the C vector (preferred observations) — the agent learns policies that produce preferred outcomes
- The **value function** maps to expected free energy — learned utilities are estimates of long-term policy quality
- **Model-free RL** (Q-learning, policy gradient) learns action-value mappings without explicitly modeling dynamics — analogous to habit learning in Active Inference
- **Model-based RL** learns a dynamics model (B matrix) and plans using it — analogous to the full Active Inference planning process

The key distinction: Active Inference provides a **principled** mechanism for balancing exploration and exploitation through the epistemic-pragmatic decomposition of EFE, while standard RL requires ad hoc exploration bonuses (ε-greedy, entropy regularization).

4. Transfer Learning and Domain Adaptation

Transfer learning enables a robot trained in one context to adapt quickly to a new one:

- **Sim-to-real transfer**: A robot trained in simulation must adapt to the "reality gap" — the discrepancy between the simulated and real dynamics. The adaptation process is parameter learning with strong structural priors from simulation.
- **Task transfer**: A robot trained to grasp one object class can adapt to novel objects by retaining the general grasping structure (motor synergies) while updating object-specific parameters.
- **Domain randomization**: Training in simulation with randomized physics parameters produces a generative model that is robust to parameter uncertainty — the model has learned to marginalize over parameter variation.

Applications

- **Self-tuning industrial robot**: An industrial robot arm that detects increasing friction in Joint 3 (persistent prediction error on Joint 3's trajectory tracking) automatically increases the control gain for that joint — compensating for wear without requiring maintenance shutdown or manual recalibration.
- **Autonomous vehicle adaptation**: A self-driving car arriving in a new city encounters unfamiliar road markings and driving conventions. Online learning updates the visual perception model (A matrix: what lane markings look like here) and behavior model (B matrix: how other drivers behave at intersections) over the first hours of operation.

Conclusion

Adaptive control — online system identification, gain scheduling, reinforcement learning, and transfer learning — implements Active Inference learning in the control domain. The robot continuously refines its generative model to maintain performance despite uncertainty and change. The next module examines communication between multiple robots in control-estimation frameworks.